

A Comparative Study of Pandas and Modin Communities

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INTRODUCTION

BACKGROUND AND MOTIVATION

Open source library and codebases form the backbone of Python Data Science ecosystem. Popular libraries such as Pandas are not only repositories, but they form a community of interactions, maintainers, and reviewers who continuously strive to make the project better. Although GitHub stars and downloads are one of metrics of a project's popularity, it relays no information about the health of the community sustaining it. If a project that has millions of active users and is used in many industrial workflows, is sustained by one/few developers, it is just a few departures away from failing entirely.

Hence, to understand community health within these repositories, one must look at reviewing responsibilities, contributor base and study the project's development discourse. For this, network analysis and natural language processing (NLP) provides tools to make these structures comparable.

The two repositories that are studied in this report are Pandas and Modin. Pandas is one of the most popular and widely-used data manipulation libraries, and Modin is a relatively new library serving as an alternative to Pandas. With them being almost in the same domain, it would be interesting to study both the projects which are marked by differing stages of maturity: with Pandas being more mature and popular, and Modin being newer.

This study would hence seek to understand measurable differences in community structure, interaction networks and topics discussed in both of these libraries.

RESEARCH QUESTION

Do contributor interaction networks and pull request topic patterns differ between a mature, widely-adopted data science library (pandas) and a smaller, newer project in the same ecosystem (Modin), and do these differences reflect meaningful variation in community health and sustainability?

PROJECT OBJECTIVES

OBJECTIVE 1: Constructing contributor interaction networks, contribution influence, and detecting communities for both pandas and Modin using pull request review data to analyse networks and relationships.

OBJECTIVE 2: Performing sentiment analysis and topic modelling in pull request text to compare the thematic focus and overall sentiment in development discourse

OBJECTIVE 3: Synthesize findings across all analysis to determine whether all of the tools provide reliable signals of community health across both of the projects.

SUCCESS CRITERIA

SUCCESS CRITERIA 1: With respect to the tools used in objective 1, metrics show visually meaningful differences between pandas and modin.

SUCCESS CRITERIA 2: With respect to objective 2, topic modelling surfaces distinct interpretable topics for each project, which allows different thematic structures (bug vs fixes etc). Sentiment analysis from pull requests displays the overall sentiment of the reviewers.

SUCCESS CRITERIA 3: In line with objective 3, the amalgamation of all the tools and techniques, a succinct and interpretable answer to the research question is formed. Confirming that the observed differences reflect different community health or not.

DATA COLLECTION AND PREPARATION

DATA SOURCES AND DATASET DESCRIPTION

Data was collected through GitHub’s API through a personal access token. Two repositories (pandas and modin) were selected.

For each repository, 2 different datasets were collected through API calls:

DATASET 1:

This is the pull request (PR) metadata. A total of 1000 recent PRs were collected for each repository. It contains basic information regarding the PR such as title, author etc. The column “body”, which is the description written by the author making the PR. This column is a primary input for NLP, which would help assess the topics the topics that are prevalent.

It contains the following fields:

Feature	Description
pr_id	Unique identifier of PR
pr_title	Title given to the PR
pr_author	Individual submitting the PR
state	Final outcome – whether closed or merged
created_at	Timestamp of PR creation
merged_at	Timestamp of PR merging
Body	Description written by author

Table 1: Dataset 1 Features

DATASET 2:

The second dataset contains PR review information. Since the basic metadata API call does not have ‘reviewed by’ relationship, a separate API call needs to be made by providing the “pr_id”. To adhere to API call limits, reviewer relationship for the most recent 300 PRs were selected. This ensured that the data collection is done within a single call, and is not time consuming. The 300 mark is also appropriate for network analysis. This dataset contains features such as “reviewed_by” which can serve as a primary input for network analysis.

Feature	Description
pr_id	Unique identifier of PR
pr_author	Individual submitting the PR
reviewed_by	Individual reviewing the PR
state	Final outcome – whether closed or merged
review_text	Text body of the review, mostly empty

DATA CLEANING AND PREPROCESSING

The collected pull request and reviews datasets from python Pandas and Modin repositories required pre-processing before network and NLP analysis could be performed. Since GitHub discussions may contain missing values, duplicated records, inconsistent text formatting and irrelevant textual elements, preprocessing was applied to improve data quality and reduce noise.

Initially missing values were examined across important variables including body, review_text and fields with timestamps. Missing textual values were replaced with empty strings to preserve records during text processing, which missing merge dates were retained as they might indicate pull requests that were closed without being merged. Duplicate records were removed to avoid repeated interactions affected later analyses such as contributor relationship and topic frequencies. Timestamp data (creation date, merging date and review date) are converted into a standard datetime format for temporal analysis.

For NLP processing pull request titles, descriptions and review comments were combined into a single text representation to capture richer discussion context. The text was then pre-processed by converting all words to lowercase, removing URLs, punctuation, and unnecessary whitespace, followed by stopwords removal. Lemmatization was applied to reduce words to their base forms while preserving semantic meaning.

Finally, additional variables such as merge duration and contributor interaction frequencies were generated to support later network and sustainability analysis.

LIMITATIONS AND DATA CONSTRAINTS

Firstly, there is considerable bot activity in Github to automate PR responses. They interact with a lot of PRs and can appear central nodes in network analysis, hence they are removed. Secondly, PR descriptions, which are used for sentiment analysis and topic modelling, are very technical in nature. They also vary in considerably in length and content, which can hamper topic modelling and sentiment analysis.

Furthermore, the data was collected through GitHub API, which only displays interaction available on GitHub. However, contribution interactions on slack, emails etc are not reflected. This ultimately means that the contribution network is limited to GitHub workspace and is not representative of the overall network.

Another limitation is that a total of 300 PRs were selected for each repository. Although for Pandas that is a very small sample size, and it covers the timeframe from only Feb 2026 to May 2026. However, for Modin the sample size is now greater, and reflects a year worth of PRs. Hence, this difference should be taken into account.

UNDERSTANDING COLLABORATION STRUCTURES

The purpose of this section is to examine how contributors interact within the Pandas and Modin communities and determine whether differences in collaboration patterns may provide insights into community organisation and sustainability. Rather than analysing contributors individually, network analysis was used to represent interactions as a structured graph, enabling the identification of highly connected users and overall collaboration behaviour.

Network visualisation was then used to explore the overall structure of contributor relationships and identify important contributors within the network. Node size was scaled according to degree centrality, where larger nodes indicate contributors with higher interaction activity.

CONSTRUCTING THE CONTRIBUTION NETWORK

Contributor interaction networks were constructed using pull request review information extracted from Pandas and Modin GitHub repositories. Contributors are presented as nodes, while Directed edges represent interactions from reviewers to pull request authors because contributor relationships are not necessarily reciprocal. Direction therefore preserves the flow of interaction and more accurately captures collaboration behaviour. In order to have a better understanding of collaboration patterns within each project, network visualisations were generated to identify highly connected contributors and examine the overall interaction structure.

The network was modelled as a directed and weighted graph. A directed structure was selected because interactions occur from reviewers toward pull request authors and therefore contain interaction direction. Edge weights represent the frequency of interactions between contributors, where larger weights indicate repeated review relationships. Using weighted edges allows stronger contributor relationships to be distinguished from occasional interactions.

Several preprocessing decisions were applied before constructing the network. Duplicate interaction records were removed to avoid repeated relationships, missing contributor values were excluded, and bot accounts were

removed to ensure the network represented human contributor interactions only. Interaction frequencies between contributors were aggregated to generate edge weights and reduce noise from repeated review events.

PANDAS CONTRIBUTION NETWORK

Figure 1: Contributor collaboration network for Pandas

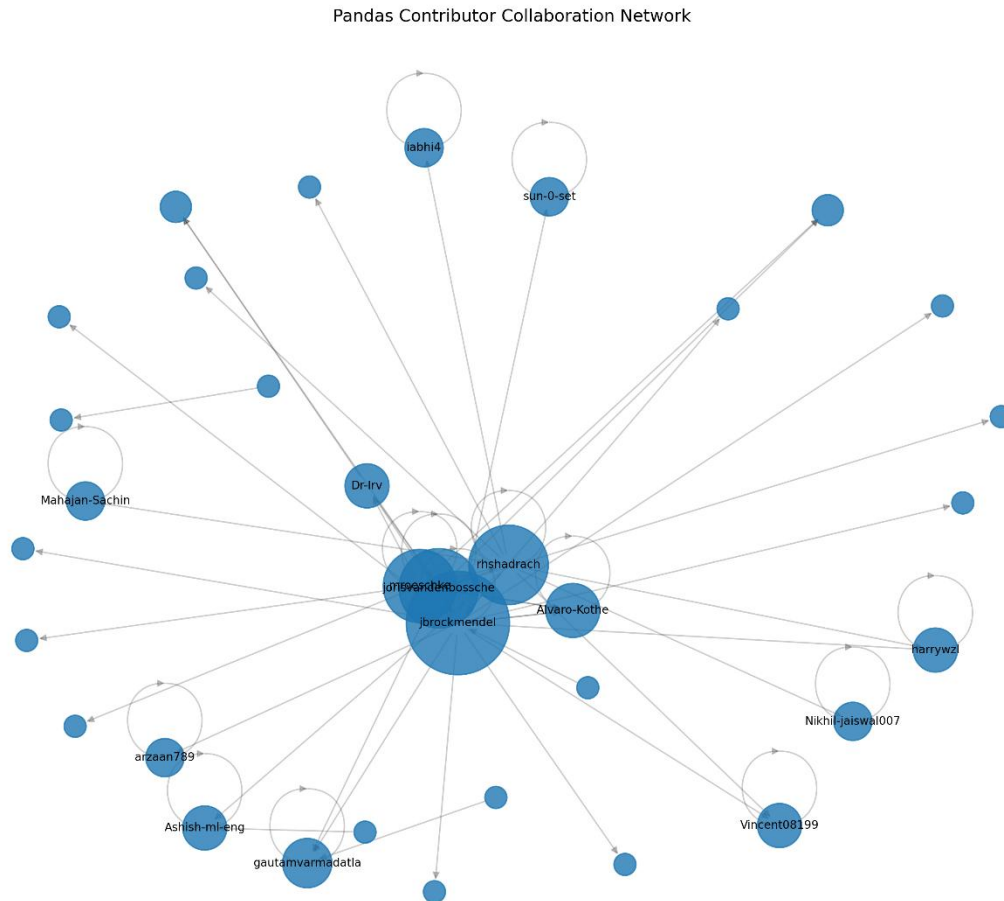


Figure 1 illustrates a Pandas contributor interaction network consists of **34 nodes** and **61 edges**, with a network density of **0.0544** and an average degree of **3.59**. The relatively low density suggests that collaboration is not evenly distributed across contributors and instead occurs through specific interaction pathways.

The network exhibits a **hub-like structure**, where a small number of contributors occupy central positions. Contributors such as **jbrockmendel**, **rhshadrach**, and **Alvaro-Kothe** appear to be more highly connected, suggesting important roles in coordinating reviews and development activities. In contrast, contributors including **MahajanSachin**, **harrywz**, **Vincent08199**, and **sun-0-set** appear at the network periphery with fewer interactions, potentially representing occasional contributors. Overall, the network indicates that Pandas collaboration relies on a smaller group of highly active contributors acting as central coordination points.

MODIN CONTRIBUTION NETWORK

Figure 2: Contributor collaboration network for Modin

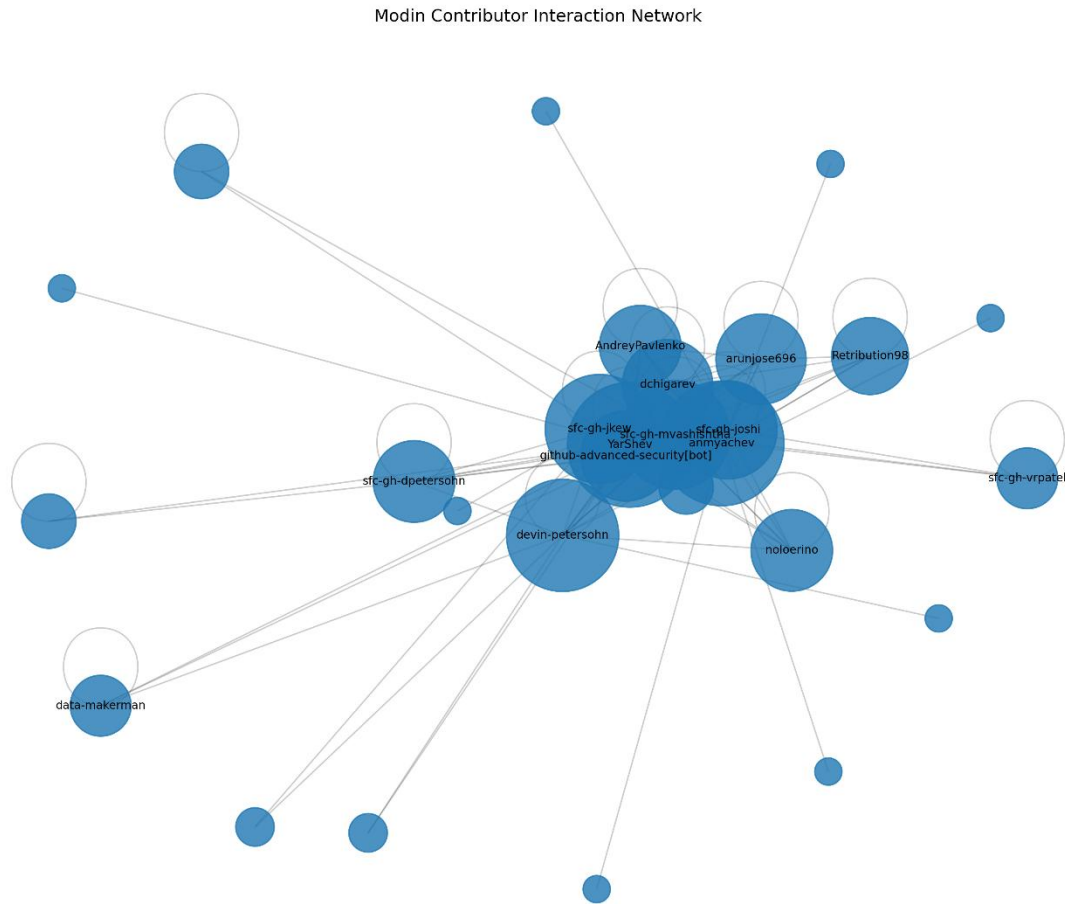


Figure 2 illustrates Modin contributor interaction network consists of 28 nodes and 103 edges, with a network density of 0.1362 and an average degree of 7.36. Compared with Pandas, the higher density and average degree indicate a more interconnected collaboration structure where contributors interact more frequently with one another.

The network demonstrates a concentrated but collaborative structure, with contributors such as YarShev, anmyachev, devin-petersohn, sfc-gh-jkew, and sfc-gh-mvashishtha occupying central positions within the network. Their higher degree centrality values suggest they play significant roles in coordinating contributor interactions and project development activities. In contrast, contributors including data-makerman and sfc-gh-vrpatel appear at the network periphery with fewer interactions, indicating lower participation levels within review activities.

Overall, the network suggests that Modin exhibits stronger interaction connectivity among contributors compared with Pandas, potentially reflecting a more closely connected development community despite involving fewer contributors.

MEASURING CONTRIBUTOR INFLUENCE

To further understand collaboration dynamics within the contributor networks, influence was measured using network centrality metrics. Degree centrality, betweenness centrality, and closeness centrality were selected as they capture different aspects of contributor importance within the network structure. Degree centrality identifies contributors with a high number of direct interactions, betweenness centrality highlights contributors who may

act as bridges connecting different users or groups, and closeness centrality measures how efficiently a contributor can reach others within the network. Using multiple measures provides a broader understanding of contributor roles, as influence cannot always be represented by interaction frequency alone. This approach enables the identification of highly active contributors as well as contributors who play important coordination and information flow roles within the community.

PANDAS FRAMEWORK:

Figure 3: Influence Metrics for Top Pandas Contributors

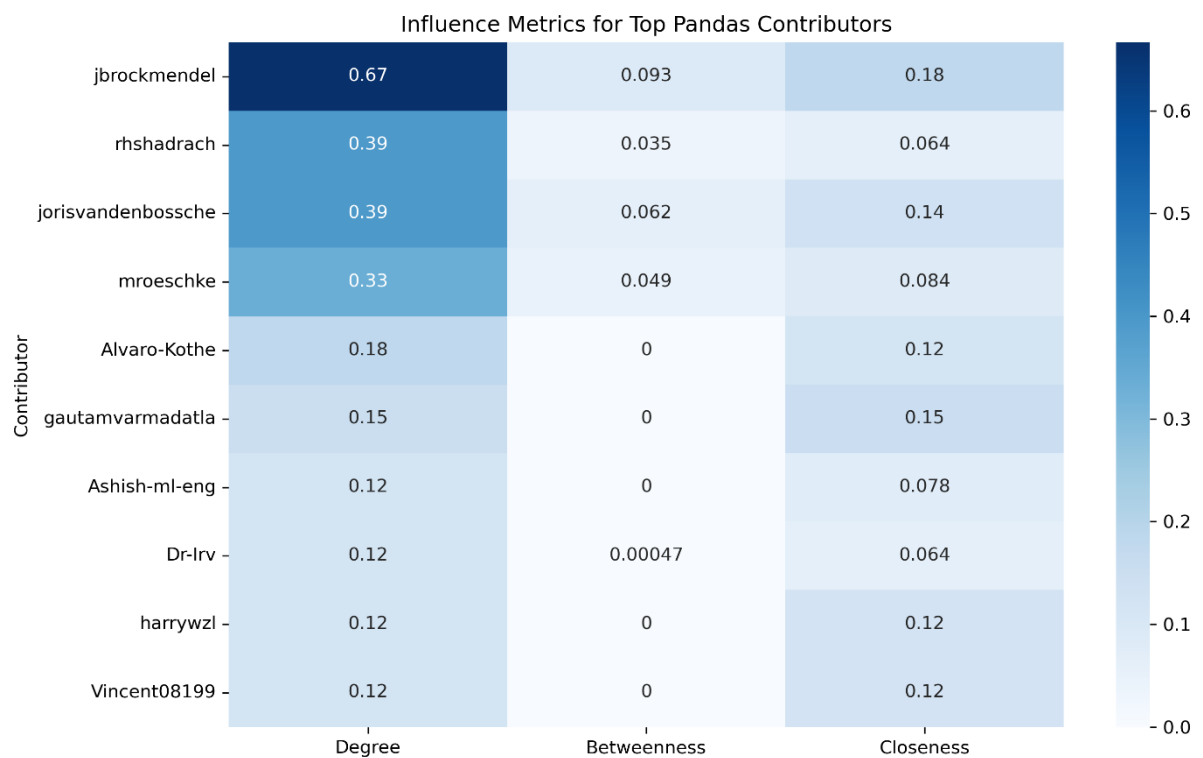
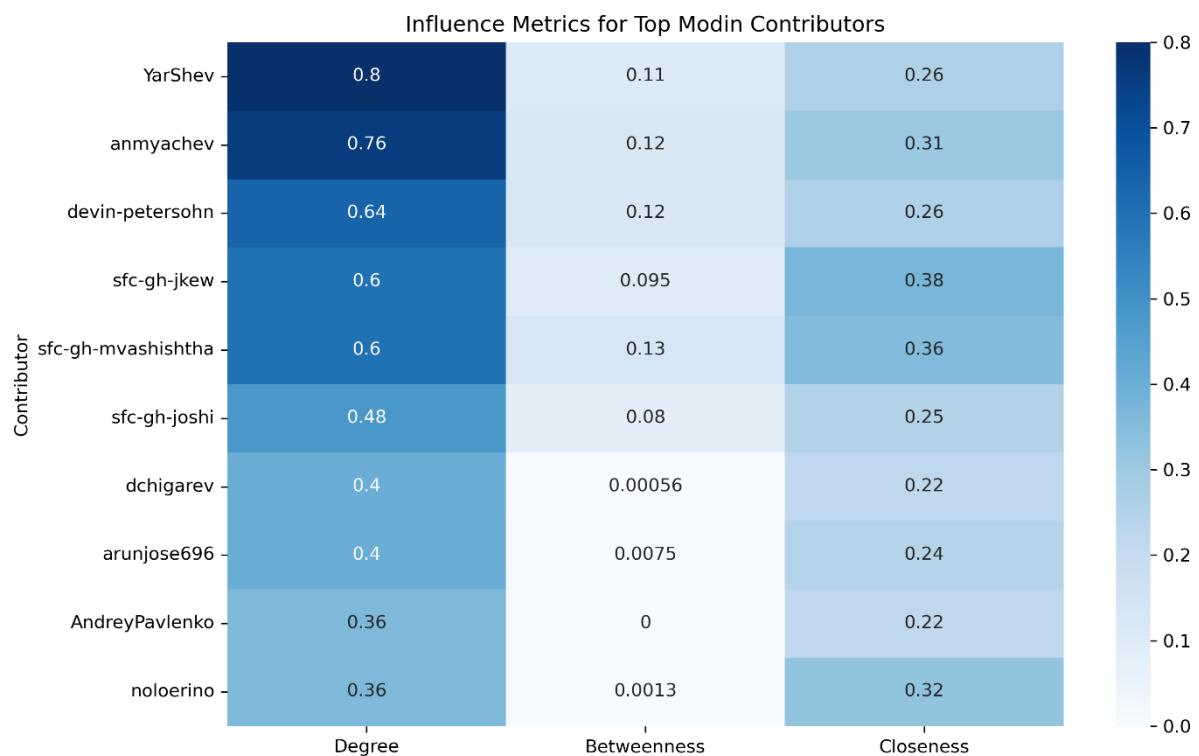


Figure 3 indicates influence analysis demonstrates that contributor importance within the Pandas network is unevenly distributed. **jbrockmendel** exhibits the highest degree centrality (**0.67**), indicating interaction with a larger proportion of contributors compared with other users. This contributor also records the highest betweenness centrality (**0.093**), suggesting an important role in connecting different contributors and facilitating information flow within the network. Contributors such as **rhshadrach**, **jorisevandenbossche**, and **mroeschke** also display relatively high degree values, indicating active participation in contributor interactions. In contrast, contributors including **Ashish-ml-eng**, **harrywzl**, and **Vincent08199** demonstrate lower influence scores, suggesting more limited participation within collaboration activities. Overall, the results indicate that the Pandas community relies on a smaller number of highly influential contributors who occupy central positions within the network structure.

MODIN FRAMEWORK:

Figure 4: Influence Metrics for Top Modin Contributors



The influence analysis of the Modin contributor network indicates that contributor activity is concentrated among a smaller group of highly connected users. **YarShev** demonstrates the highest degree centrality (**0.80**), indicating interaction with a large proportion of contributors within the network. Contributors such as **anmyachev**, **devin-petersohn**, **sfc-gh-jkew**, and **sfc-gh-mvashishtha** also exhibit relatively high degree values, suggesting active participation in project collaboration activities. Furthermore, **sfc-gh-mvashishtha**, **anmyachev**, and **devin-petersohn** display comparatively higher betweenness values, indicating potential roles as bridges facilitating communication between contributors. Contributors such as **AndreyPavlenko** and **noloerino** show lower overall influence scores and appear to play lesser central roles within the interaction network. Overall, the results suggest that although Modin contains fewer contributors than Pandas, collaboration activities are more concentrated around a group of influential contributors who help maintain communication and coordination within the community.

DETECTING COMMUNITIES AND INTERACTION PATTERNS

Beyond measuring individual contributor influence, it is also important to examine whether contributors form collaboration groups within the network. Community detection analysis was used to identify clusters of contributors who interact more frequently with each other than with the broader network. Analysing these interaction patterns helps reveal how development activities are organised and provides insights into collaboration behaviour and community sustainability.

COMMUNITY DETECTION CLUSTER (PANDAS)

Figure 5: Contributor Interaction Clusters in Pandas

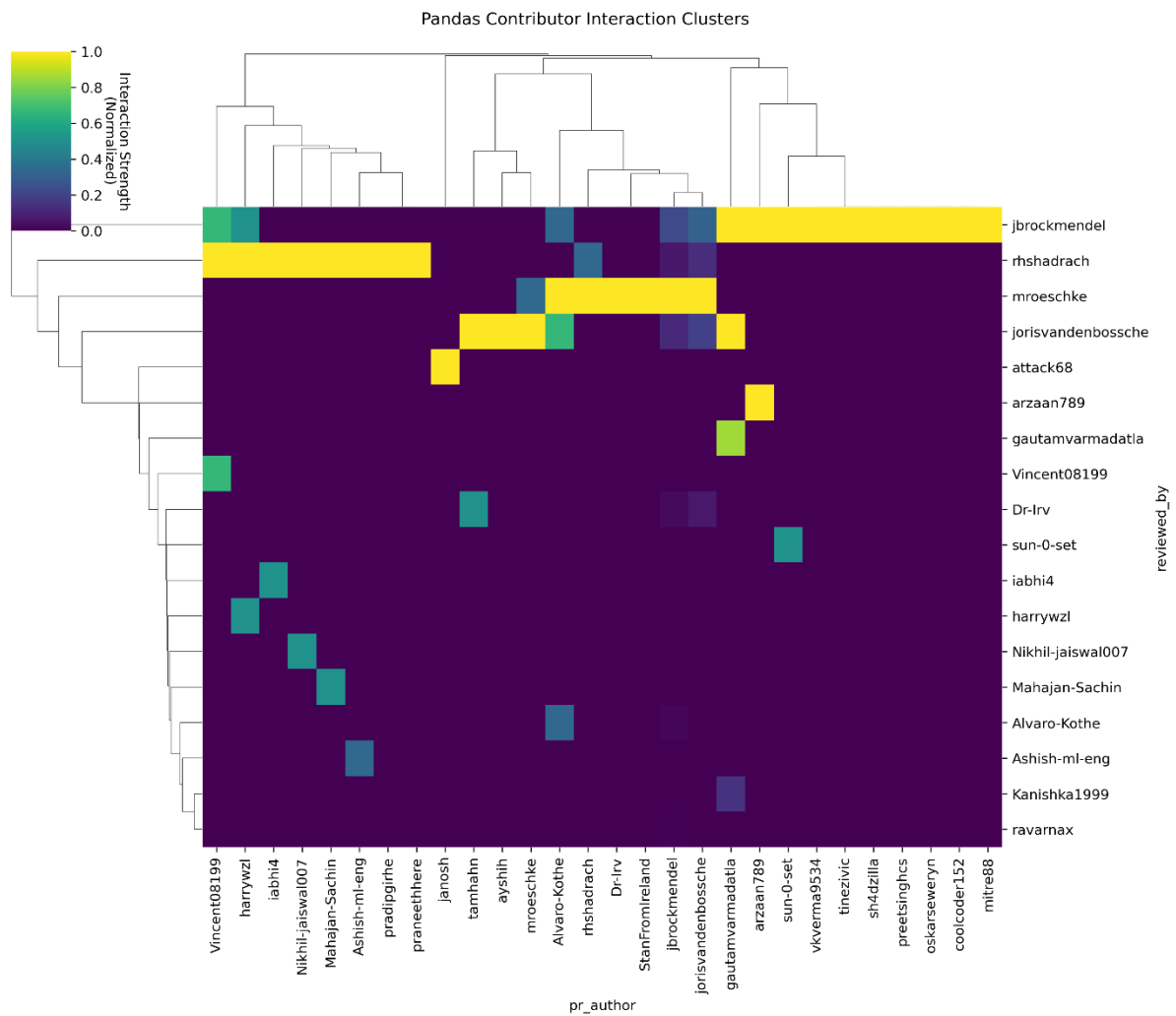
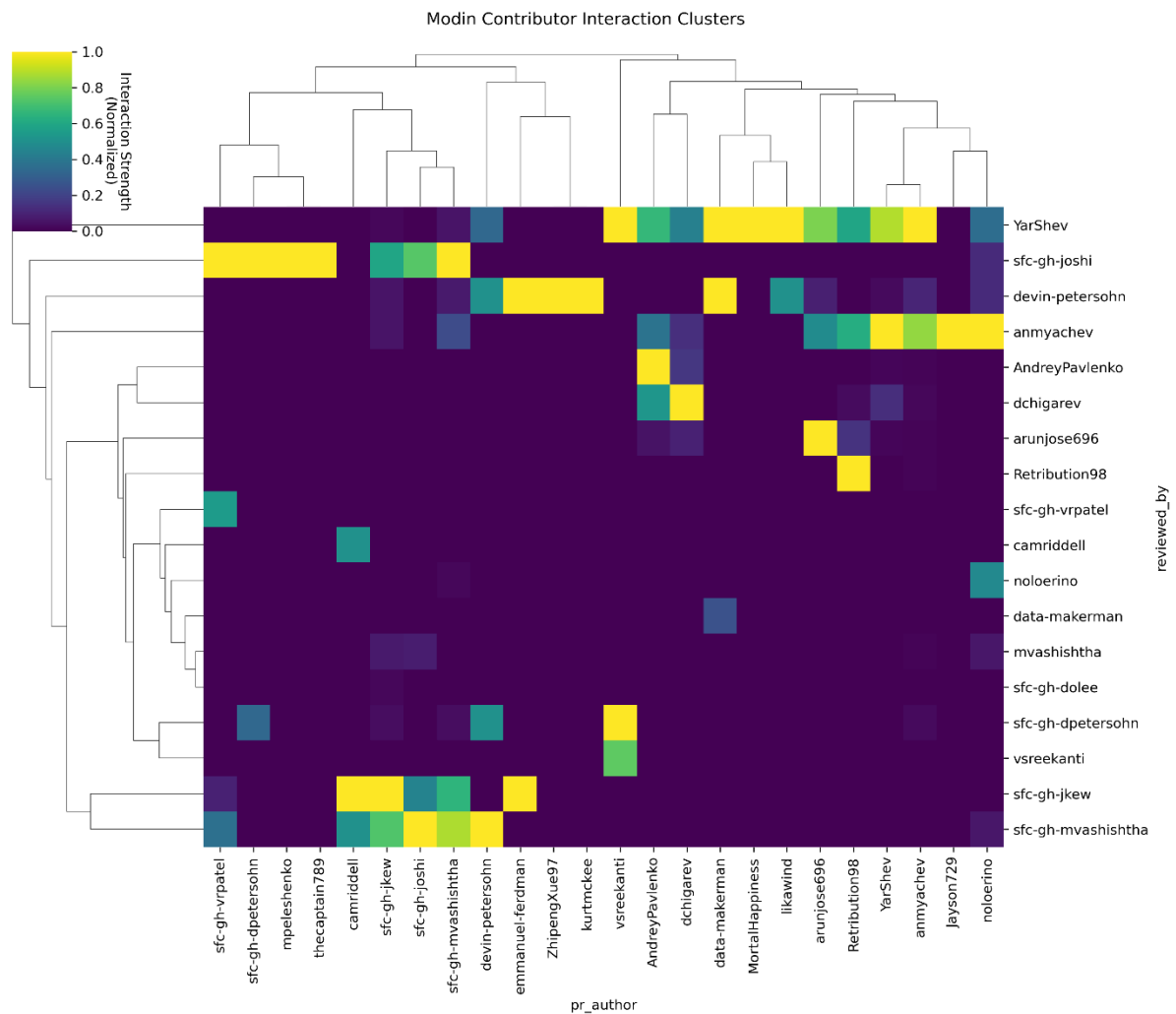


Figure 5 is a cluster map that reveals distinct interaction patterns among contributors within the Pandas project. Contributors such as jbrockmendel, mroeschke, rhshadrach, and jorisvandenbossche appear within stronger interaction clusters, suggesting more frequent collaboration behaviour. The dendrogram structure indicates the presence of multiple contributor groups rather than a single highly connected community. In contrast, several contributors remain relatively isolated with weaker interaction patterns. This suggests that while Pandas relies on a set of highly active contributors, collaboration activity is distributed across smaller interaction communities.

COMMUNITY DETECTION GRAPH (MODIN)

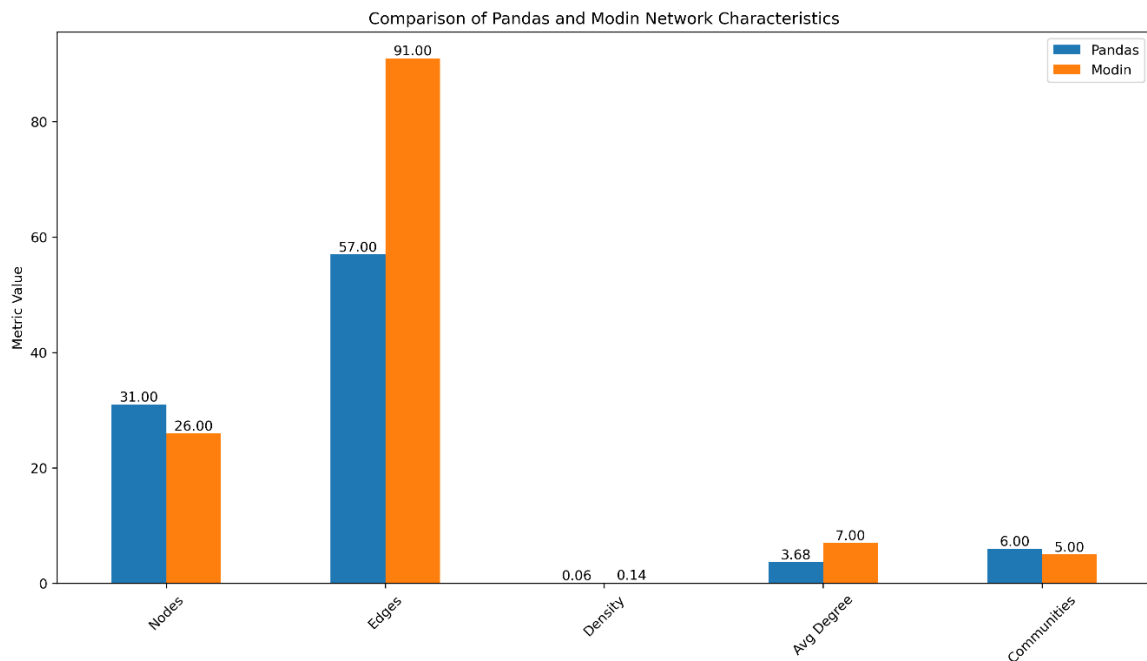
Figure 6: Contributor Interaction Clusters in Modin



The contributor cluster map for Modin reveals stronger and more concentrated interaction patterns compared with Pandas. Contributors such as YarShev, anmyachev, devin-petersohn, sfc-gh-jkew, and sfc-gh-mvashishtha appear within stronger interaction regions, indicating frequent collaboration activity among these contributors. The dendrogram structure suggests the presence of several contributor groups with closer interaction behaviour, while fewer contributors appear isolated compared with the Pandas network. In contrast, contributors such as data-makerman and noloerino demonstrate weaker interaction patterns and appear more peripheral within the collaboration structure. Overall, the clustering results suggest that Modin exhibits a more interconnected and tightly coordinated contributor community despite involving fewer contributors than Pandas.

COMPARATIVE ANALYSIS OF COLLABORATIVE STRUCTURES

Figure 7: comparison of Pandas and Modin Network characteristics



The comparative analysis highlights structural differences between the Pandas and Modin contributor communities. Pandas exhibits a larger contributor base (31 nodes) and a greater number of detected communities (6), suggesting a broader but more distributed collaboration structure. In contrast, Modin demonstrates a higher number of interaction edges (91), greater network density (0.14), and a higher average degree (7.00) compared with Pandas (3.68). These findings indicate that although Modin contains fewer contributors, interactions between contributors are more concentrated and interconnected. Such differences suggest that mature and newer projects may adopt different collaboration patterns, with Pandas relying on a wider contributor base and Modin exhibiting tighter coordination among contributors.

COLLABORATION INSIGHTS

The summary comparison further highlights differences in collaboration characteristics between the two projects. Pandas contains a larger number of contributors and communities, indicating broader participation across the project. However, Modin demonstrates higher network density, a greater number of interaction edges, and a higher average degree value, suggesting stronger contributor interconnectedness. These findings imply that although Pandas benefits from a wider contributor base, Modin exhibits more tightly coordinated interaction patterns. Such differences may reflect varying collaboration strategies between mature and emerging open-source communities and provide insight into factors related to community sustainability.

EXPLORING DEVELOPMENT DISCUSSIONS

The purpose of this section is to examine the textual content within pull request discussions and review comments to better understand contributor behaviour and development activities within the Pandas and Modin communities. While network analysis provides insights into contributor relationships, textual analysis enables the identification of discussion patterns, contributor sentiment, and recurring development themes. Natural Language Processing (NLP) techniques were applied to extract meaningful information from pull request descriptions and review interactions in order to identify differences in communication behaviour and project development focus.

NLP FRAMEWORK AND ANALYTICAL APPROACH

To analyse development discussions, Natural Language Processing (NLP) techniques were applied to pull request descriptions and review comments from the Pandas and Modin repositories. TF-IDF was used to identify important discussion terms, sentiment analysis evaluated interaction tone, topic modelling identified hidden themes, and temporal analysis examined contribution patterns over time. Together, these methods provide insights into contributor behaviour and potential indicators of community sustainability.

TEXT FREQUENCY ANALYSIS AND TF-IDF TERM EXTRACTION

Figure 8: Comparison of important discussion terms in Pandas and Modin

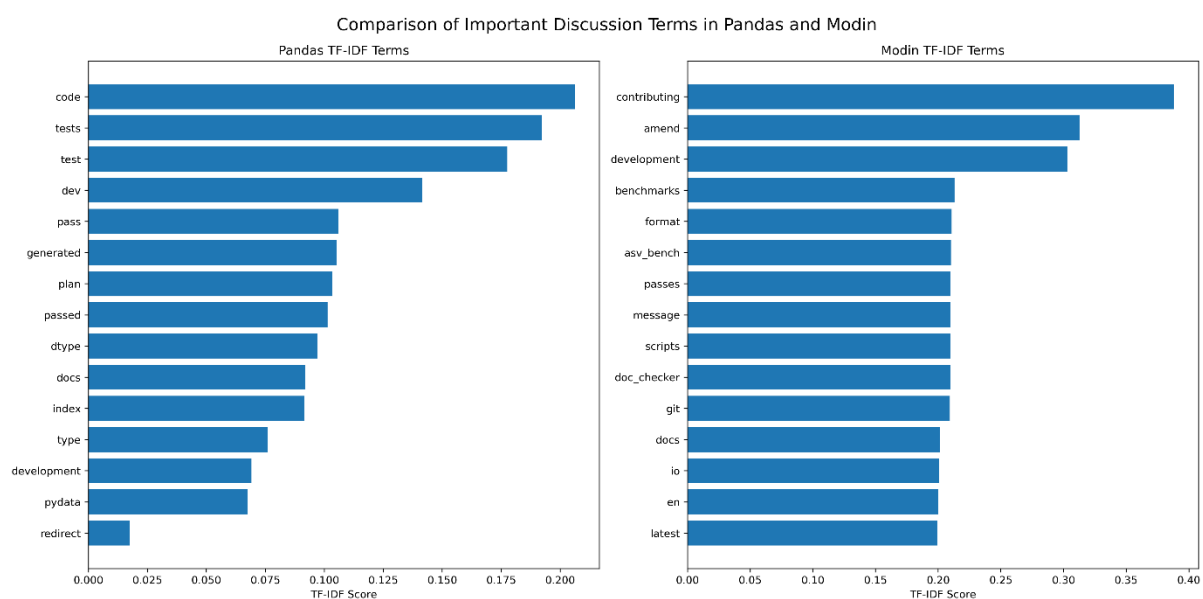


Figure 8 compares important discussion terms extracted from Pandas and Modin pull request conversations using TF-IDF analysis. The Pandas project shows greater emphasis on terms related to code implementation, testing procedures, indexing, and data structures, suggesting discussions primarily focused on maintaining functionality and code quality. In contrast, Modin discussions contain terms associated with benchmarking, contribution processes, and development activities, indicating a stronger focus on performance optimisation and workflow management. These differences suggest variation in development priorities between the two projects despite operating within a similar data science ecosystem.

SENTIMENT ANALYSIS OF PULL REQUEST DISCUSSIONS

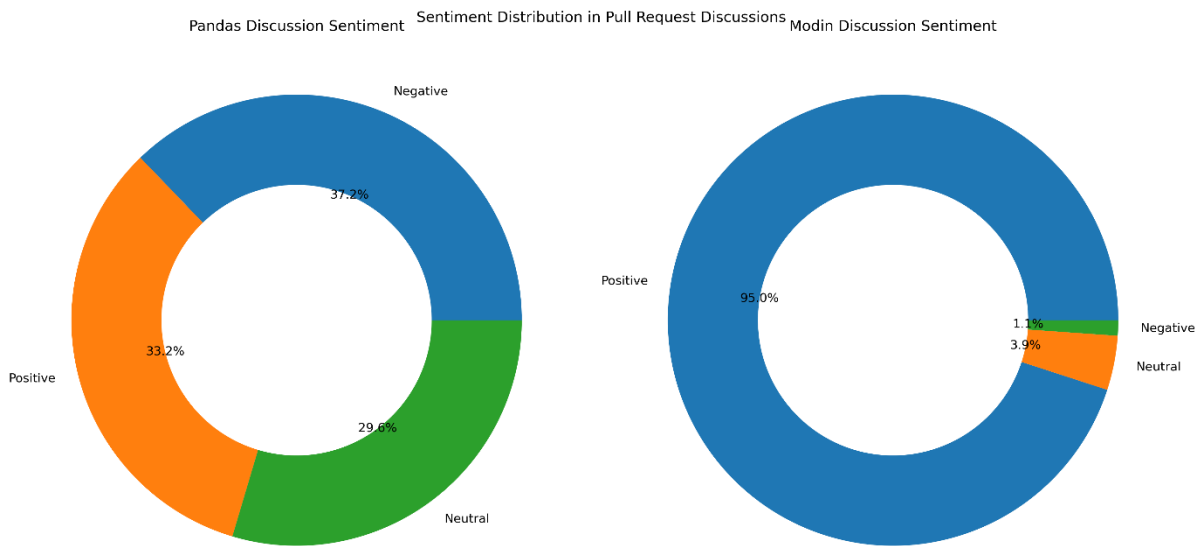
Sentiment analysis was performed to examine the overall tone of contributor discussions within pull request descriptions and review interactions. The VADER (Valence Aware Dictionary and Sentiment Reasoner) approach was applied to calculate sentiment scores and classify discussions into positive, neutral, and negative categories. Analysing discussion tone provides insight into contributor communication behaviour and may indicate differences in collaboration patterns between the Pandas and Modin communities.

VADER SENTIMENT ANALYSIS

Sentiment analysis was conducted to examine the overall tone of pull request discussions within the Pandas and Modin communities. The VADER (Valence Aware Dictionary and Sentiment Reasoner) approach was used to calculate compound sentiment scores and classify discussions into positive, neutral, and negative categories. VADER thresholds of ≥ 0.05 for positive, ≤ -0.05 for negative, and values between -0.05 and 0.05 for neutral

sentiment were adopted to ensure consistent classification. Analysing discussion tone provides insights into contributor communication behaviour and may reveal differences in collaboration dynamics between the two projects.

Figure 9: Sentiment Distribution in Pull Request Discussions Across Pandas and Modin



Although VADER provides a simple method for measuring discussion sentiment, it was originally designed for social media and general language rather than technical software discussions. Pull request conversations often contain domain-specific terminology and limited emotional expression. Consequently, terms such as **"fixed"**, **"passed"**, or **"failed"** may influence sentiment scores despite not necessarily representing contributor attitudes. Therefore, sentiment results should be interpreted cautiously and considered as indicators of discussion tone rather than direct measures of contributor emotion.

POLARITY DISTRIBUTION

Figure 10: Polarity Distribution of Pull Request Discussions

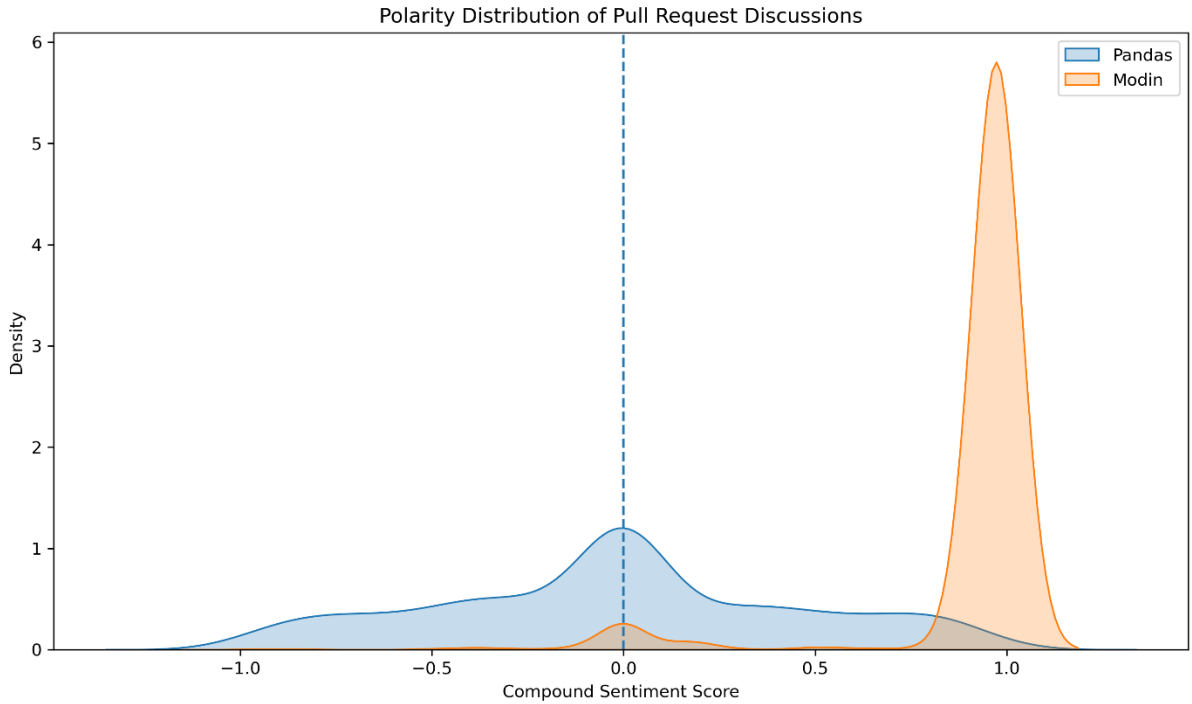


Figure 10 shows the polarity distribution of pull request discussions for Pandas and Modin using VADER compound sentiment scores. The dashed line at 0 represents neutral sentiment, with values on the left indicating negative sentiment and values on the right indicating positive sentiment. Pandas displays a wider spread of sentiment scores, suggesting more varied discussion tones. In contrast, Modin shows a strong concentration toward positive values, indicating more consistently positive discussion patterns. However, since pull request discussions contain mainly technical language, terms such as *fixed*, *passed*, or *added* may influence sentiment scores despite not reflecting actual contributor emotions.

TOPIC MODELLING AND DISCUSSION THEMES

Topic modelling was applied to identify hidden discussion themes within pull request conversations from the Pandas and Modin repositories. Unlike TF-IDF, which identifies important individual terms, topic modelling groups related words together to reveal broader discussion patterns. Latent Dirichlet Allocation (LDA) was used to uncover underlying themes and provide insight into common development activities and contributor focus areas.

LDA TOPIC MODELLING APPROACH

Latent Dirichlet Allocation (LDA) was applied to identify hidden discussion themes within pull request descriptions and review comments from the Pandas and Modin repositories. Prior to modelling, standard English stopwords and project-specific custom stopwords were removed, along with URLs and repository-related terms that do not contribute meaningful semantic information. The cleaned text was transformed into a document-term matrix and used as input for topic extraction.

TOPIC NUMBER EVALUATION

The x-axis represents the number of topics specified for the LDA model, while the y-axis represents the perplexity score used to evaluate model performance. Lower perplexity values indicate a better fit between the model and the discussion data, suggesting more meaningful topic extraction.

Figure 11: LDA Topic Number Evaluation

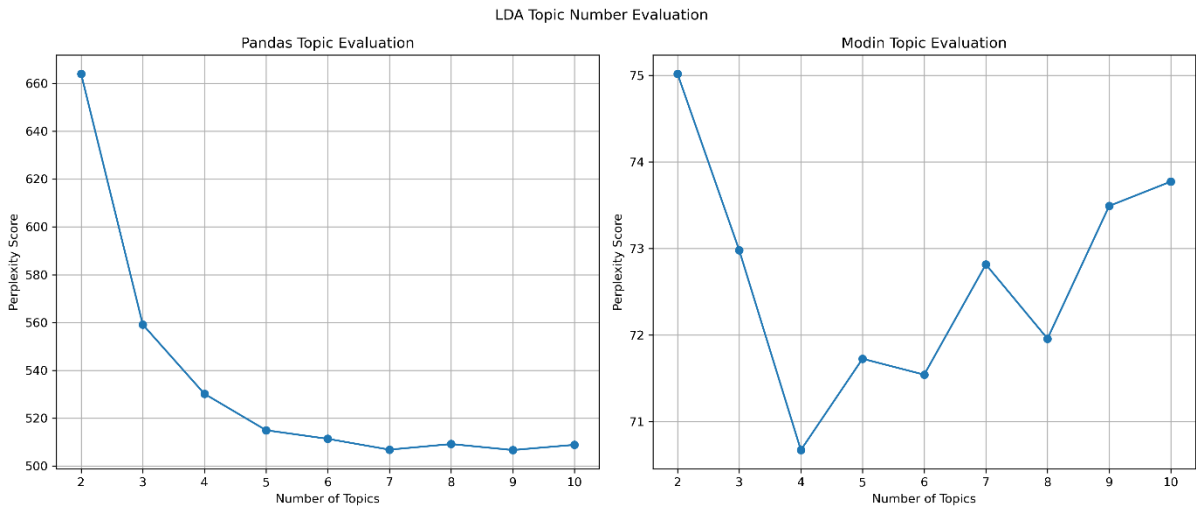


Figure 11 presents the LDA topic number evaluation for the Pandas and Modin repositories using perplexity scores. Lower perplexity values indicate improved topic model performance. For Pandas, the perplexity score decreases substantially before stabilising around 6–7 topics, suggesting limited improvement beyond this point. For Modin, the lowest perplexity score occurs at approximately 4 topics before increasing again. Based on these results, topic numbers were selected to balance model performance and topic interpretability.

EXTRACTED TOPICS:

Figure 12: LDA Topic themes

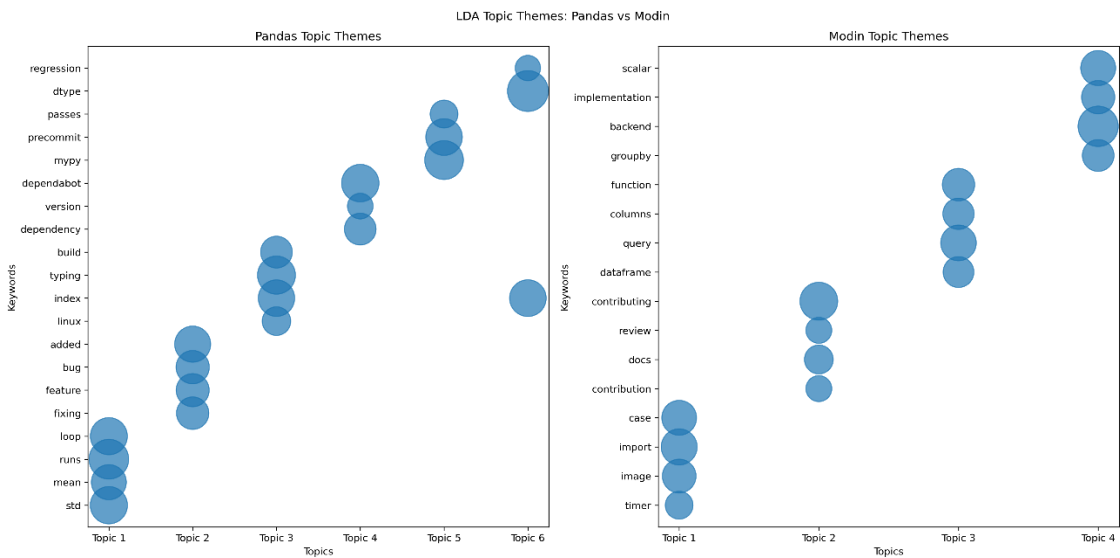


Figure 12 presents the topic themes extracted from pull request discussions using LDA topic modelling for the Pandas and Modin repositories. Each bubble represents an important keyword associated with a specific topic, while larger bubbles indicate greater contribution of a keyword within that topic. The Pandas repository exhibited themes related to performance analysis, bug fixing, dependency management, testing procedures, and data

structure implementation. In contrast, Modin discussions focused more on contributor workflows, dataframe operations, functionality issues, and backend implementation. These results suggest differences in development priorities and discussion patterns between the two projects.

TEMPORAL ANALYSIS OF DISCUSSION AND CONTRIBUTION ACTIVITY OVER TIME

Figure 13: Contribution activity

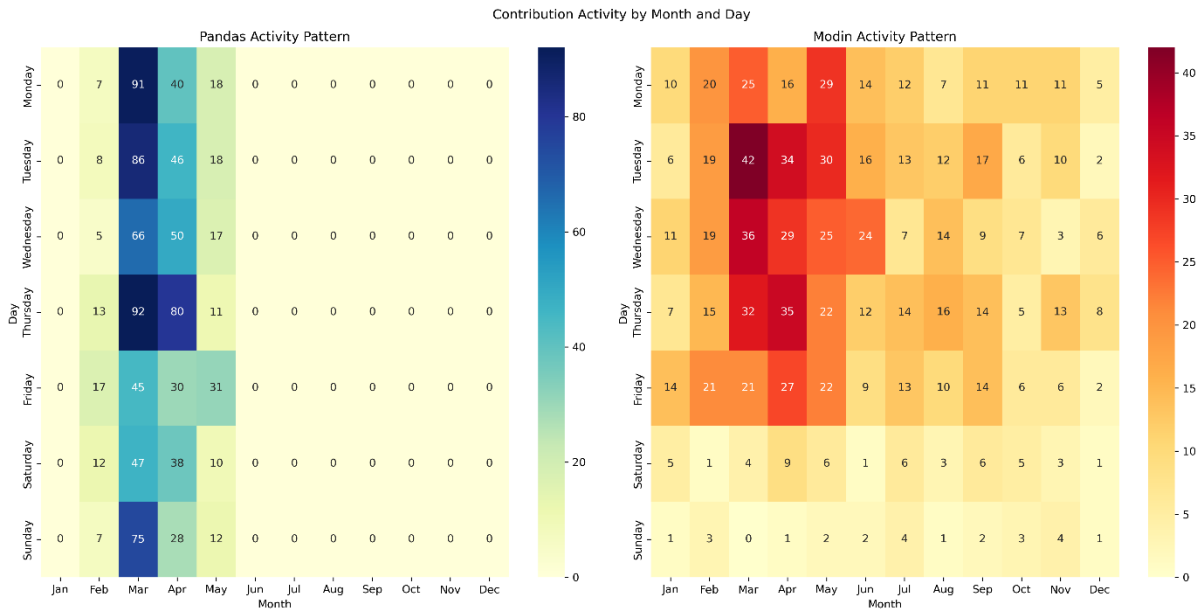


Figure 13 presents aggregated contribution activity patterns across months and days of the week for the Pandas and Modin repositories. Darker regions indicate higher pull request activity levels. Pandas contributions are concentrated within a shorter period, primarily between March and May, with higher activity observed on weekdays. In contrast, Modin exhibits a more distributed activity pattern across multiple months, suggesting a more consistent contribution process over time. Both repositories show lower activity during weekends compared to weekdays.

CONCLUSION

This study examined whether contributor interaction networks and pull request discussion patterns differ between Pandas and Modin, and whether such differences reflect meaningful variation in community health and sustainability. Drawing on network analysis, sentiment analysis, and topic modelling, the findings confirm that the two communities exhibit structurally and thematically distinct patterns consistent with their differing stages of maturity.

Network analysis revealed that Pandas operates with a broader contributor base distributed across more detected communities, though its low network density and hub-like structure indicate that collaboration is channelled through a small number of central contributors. Modin, despite having fewer contributors, demonstrated higher network density and tighter clustering, reflecting the closer coordination typical of an earlier-stage project. Influence analysis reinforced this, showing that both repositories rely heavily on a small group of highly connected individuals, with this concentration being more pronounced in Modin.

Textual analysis further distinguished the two communities. Pandas discussions centred on testing, bug fixing, and data structure maintenance, consistent with the upkeep demands of a mature library. Modin discussions

emphasised benchmarking, contributor workflows, and backend implementation, reflecting a project still actively expanding its capabilities. Sentiment analysis indicated broadly positive discussion tones across both repositories. Together, these findings suggest that the observed differences do reflect meaningful variation in community organisation. Pandas demonstrates the scale of a mature project but shows contribution concentration that poses sustainability risks. Modin exhibits strong internal cohesion but remains vulnerable due to its small contributor base.

Key limitations include the restriction of data to GitHub activity, the asymmetric temporal coverage between the two samples, and the unsuitability of VADER for technical discourse. Future work should incorporate additional communication channels, extend the sample window, and apply NLP models better suited to software engineering text.